

CURRICULUM VITAE

Dr. VAIDYANATHAN SIVAKUMAR, Ph.D.,

Associate Professor

Department of Chemistry

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Summary at IIT Hyderabad: (Oct 2022 – till date)

S.No			
1	Courses	12 (9 + 3 ongoing)	Average feedback (9 courses) – 4.4/5
2	PhD Guidance	Ongoing (Oct 2022 onwards)	7 (4 CSIR + 1 DST + 2 MoE)
		Graduated (13)	Oct 2022-till date – 7 graduated Sep 2011- Sep 2022 – 6 graduated
3	MSc Project students	Ongoing	Oct 2022-till date 2 ongoing
		Completed	Oct 2022 – till date - 5 completed Sep 2011- Sep 2022 – 22 completed
4	Internship	Completed	Oct 2022 – till date – ~18 completed Sep 2011- Sep 2022 – 20 completed
5	Publications	Total = 125 (peer review journals), Citations – 3975 h-index – 38, i10-index - 86	Oct 2022 – till date – 33 (with students – 16 (12 Q1 + 4 Q2), with no students – 17 (13 Q1 + 4Q2)) Sep 2011- Sep 2022 – 92 (38 Q1 + 47 Q2)
6	Patent	Submitted - 3	Oct 2022 – till date
7	Projects	Ongoing – 3 (~94 Lakhs)	DST, BRNS, IITH
		Completed –7 (195.75 Lakhs)	DST, Inspire, SERB, BRNS, TEQIP
		Submitted – 3 (~333 Lakhs)	ARG, DST, CSIR
		Student & Faculty exchange	IITH-NIMS (ongoing) Lotus – 2 (Submitted)
8	Collaboration established	International	Prof. NAKANISHI Takashi, NIMS, Japan Prof. C. Adachi, Kyushu University, Japan Prof. Vinich Promarak Vidyasirimedhi Institute of Science and Technology, Thailand
		National	Prof. Sabita Patel, NIT Rourkela Prof. Mahesh Raava, SRM-AP University
9	Conference/Workshop	Co-convenor	3rd International Conference on Main-group Molecules to Materials (MMM III) – Dec 9-11, 2023
		Convenor - Workshop	Smart Functional Materials 2024

Academic Profile: **Ph.D.** (Inorganic/Materials Chemistry) * – March 2007
Indian Institute of Technology (IIT) Madras, India.
Master of Science (Chemistry) – 2002, I Class
Muthurangam Govt. Arts College, Vellore – University of Madras,
India.
Bachelor of Science (Chemistry) – 2000, I Class
Muthurangam Govt. Arts College, Vellore – University of Madras,
India.

* Prof. Alfred Werner Medal for **Best Thesis** in Inorganic and Analytical Chemistry

Thesis: “Investigation of Rare Earth Luminescence in Mixed Metal Oxides and Silicates”

Thesis Advisor: **Prof. U.V. Varadaraju**
Materials Science Research Centre & Department of Chemistry
Indian Institute of Technology (IIT) Madras, Chennai 600 036, India.

Teaching and Research Profile:

Associate Professor, Department of Chemistry, IIT Hyderabad, India – Oct 2022 – Till date

Affiliated faculty at Greenko School of Sustainability – Sep 2023 – Till date

Visiting Researcher (JST) – NIMS – June – July 2025

Associate Professor, Department of Chemistry, NIT Rourkela, India – Feb 2018 – Sep 2022

Assistant Professor, Department of Chemistry, NIT Rourkela, India – Sep 2011 – Feb 2018

Inspire Faculty Fellow, Department of Chemistry, NIT Rourkela, India – Jan 13 – Dec 2017

Lecturer (Adhoc), Department of Chemistry, NIT, Calicut, India – July 2010 to May 2011

CEA Postdoctoral Researcher, CEA, Saclay, France – April 2009 to April 2010

Brain Korea Postdoctoral Fellow, KAIST, South Korea – April 2007 to March 2009

CSIR Senior Research Fellow, IIT Madras, India – Jan 2006 to March 2007

Junior Research Fellow, IIT Madras, India – Jan 2004 to Dec 2005

Project Associate, IIT Madras, India – Sep 2003 – Dec 2003

Synthetic Chemist, Process development laboratory at Orchid Chemical and Pharmaceuticals Ltd., Chennai – Jun 2003 – Aug 2003

Research Fellow – Center for Electrochemical and Energy Division, Spic Science Foundation, Chennai – Oct 2002 – May 2003

Guest Lecturer, Muthurangam Govt. Arts College, Vellore – July 2002 – Sep 2002

Distinctions, Awards & Honors :

1. **One of the top 2% scientists** in the world, according to Stanford University (Career as well as single year – 2025)
2. Single crystal data submitted to the Cambridge Crystallographic Database (CCDC)
3. **Best Oral Presentation Award:** Sandhya Rani Nayak and Sivakumar Vaidyanathan*
Illuminating the Path: Molecular Design and Photophysical Insights of Blue Emitting Fluorophores for OLEDs Applications, International Conference on Luminescence and Its Applications 2023, organized by IICT Hyderabad, IIT Hyderabad, 3-6 July 2023.

4. **Best Poster Award:** Sudipta Das, Aayush Aryan, Pardha Saradhi Maram, and Sivakumar Vaidyanathan, Novel red emitting $[\text{CaIn}(\text{PO}_4)_6 : \text{Eu}^{3+}]$ phosphor for solid state lighting application, International Conference on Luminescence and its applications 2023, organized by IICT Hyderabad, IIT Hyderabad, 3-6, July 2023.
5. **SERB International Research Experience Fellowship**, supported by SERB, DST to visit and work with Prof. C. Adachi at Kyushu University, Japan
6. **ICCD appreciation** – Powder XRD pattern included in ICCD 2021
7. **Review Editor – Frontiers in Photonics – 2022 onwards**
8. **Editorial Board Member:** *Journal of Materials Science and Nanomaterials*, scientific publishers (open access)
9. **Best Poster Award:** R. Marikumar and V. Sivakumar* “Synthesis and Photoluminescence studies of Eu^{3+} doped $\text{KGd}_5\text{O}_5(\text{VO}_4)_2$ Red phosphor for solid state lighting”, NCLA-2017, 9-11th January 2017, CSIR- Indian Institute of Chemical Technology [IICT], Hyderabad.
10. **Best Poster Award:** T. Jairam and V. Sivakumar*, “Synthesis and photophysical studies of new Triphenylamine - Phenanthroline based bipolar material for phosphorescent OLED” in NCLA 2016, February 18 - 20, R.T.M. Nagpur University, Nagpur, Maharashtra
11. **Appreciation Award:** K. Aravind Babu and V. Sivakumar*, “Green emitting heteroleptic Ir(III) complexes and their photophysical study” in NCLA 2016, February 18 - 20, R.T.M. Nagpur University, Nagpur, Maharashtra – Oral presentation.
12. **Best poster award** - K. Aravind Babu and V. Sivakumar*, “Synthesis and photophysical properties of an organic light color-conversion material for white LEDs”, 5th International Conference on Luminescence and its applications, February 9-12, 2015, Bangalore.
13. **Best poster award** - S. Kasturi, Onkar Kumar das, and V. Sivakumar* “New red emitting phosphor ($\text{NaLaCaWO}_6:\text{Eu}^{3+}$) for LED applications” National conference on Luminescence and applications (NCLA) – 2014 at Jabalpur.
14. **INSPIRE Faculty Fellowship Award** – INSPIRE, DST, Govt. of India, August 2012.
15. **DAE Young Scientist Award** – DAE, Govt. of India, June 2012.
16. **Editorial Board Member in The Scientific World Journal** – May 2012 – May 2015.
17. **Fast Track Young Scientist Award** – DST, Govt. of India, Dec 2011
18. **CEA Post-doctoral fellowship** – CEA, France (April 2009 – April 2010)
19. **Brain Korea-21 program** (April 2007 – March 2009) - Post-doctoral research fellowship
20. **Best poster award** - Joint Symposium on Materials Science and Engineering for the 21st Century 2007, Hsinchu, Taiwan
21. **Prof. Alfred Werner Medal** - Best Thesis in Inorganic and Analytical Chemistry
22. **Graduate Aptitude Test for Engineering (GATE)** - 91.66%
23. **CSIR SRF Award** – Senior Research Fellowship to carry out research
24. **Student Travel Grants** received from CSIR, DST, INSA, TNSCST, and IIT Madras for attending the International Conference in San Diego, USA.
25. **State Merit Scholarship** – B. Sc and M. Sc (1997 – 2002)

Publications:

In referred International/National journals: 125 (122/3)

In conference proceedings: 65

Conference and workshop presentations: Oral – 8 (USA, France, Korea, Singapore, Poland and India),
Poster – 41 (*Details of publications are listed in Annexure “B”*)

Area of Research:**Optoelectronic/Molecular Materials for light-emitting diodes:**

Organic materials for optoelectronics applications

Donor-acceptor fluorophores for blue light-emitting diodes

Lanthanide-containing Molecular complexes, Bipolar/blue molecular materials for organic light-emitting diodes, Synthesis of pure organic room-temperature phosphorescence materials

Molecular designing of Ir dendrimers for PHOLEDs: Synthesis of highly efficient Iridium-based dendrimers for phosphorescent organic light emitting diodes (PHOLEDs),

Synthesis of new high quantum efficient fluorophore ligands (serve as a dendron for Ir-based dendrimers) - capable of hole or electron transfer

Boron contains fluorophores and their applications in LEDs.

Heavy metal atom-free pure organic room temperature phosphorescence materials

New Phosphor and Nano (quantum dots) materials for lighting/display applications:

New highly efficient rare-earth ion-activated phosphor materials for solid-state lighting (SSL), cold cathode fluorescent lamps (CCFL), and plasma display panels (PDP) applications.

RE-activated or semiconducting nanocrystals and all-inorganic perovskite quantum dots by hydrothermal/organometallic synthesis, and explore them for white light-emitting diodes (Details given in Annexure “D”)

Courses Taught / Currently Teaching:

At the undergraduate level: At IIT Hyderabad:

IC-1030- Introduction to Systematic Inorganic Chemistry and Applications

IC-40020 – Materials Science - II

CY5140-Introduction to Optical Materials and their Applications

At NIT Rourkela: CY-221 - Basic Inorganic Chemistry – I, CY1107 Chemistry Laboratory (I B. Tech), CY 1101 Chemistry, CY 1001 Chemistry (I B. Tech), EA 171 NCC-I, EA 172 NCC-II, CY 323 Basic Inorganic Chemistry (Integrated M.Sc), CY 523 Solid State Chemistry, SHG 201 Physical Chemistry (II B. Tech), CY1094 Chemistry Laboratory (I B. Tech), SHG 291 Physical Chemistry Laboratory (II B. Tech), Inorganic Chemistry (I B.Sc)

At the postgraduate level:

At IIT Hyderabad:

CY5110 Concepts in Inorganic Chemistry

CY5140-Introduction to Optical Materials and their Applications

CY6580- Data Analysis in Inorganic Chemistry

ET5030 - Non-Conventional Energy Sources and Environment

At NIT Rourkela:

CY4202 Advanced Topic in Inorganic Chemistry, CY4201 Main Group and Organometallic Chemistry, CY4702 Inorganic Chemistry Laboratory, CY2201 Principles of Inorganic Chemistry, CY 537 Advanced Solid-State Chemistry, CY 422 Advanced Inorganic Chemistry, CY 476 Analytical Chemistry Laboratory, CY 622 Comprehensive Viva-Voce, CY 537 Special Topics Inorganic Chemistry, CY 622 Organometallic Compounds and their Application, CY 512 Chemistry of Transition and Non-transition Elements, CY 611 Advanced Analytical Chemistry, CY 572 Inorganic Laboratory, CY 6112 Inorganic Chemistry II (I M. Sc Tech), CY7103 Spectroscopy I (Electronic spectroscopy) (II M.Sc Tech), Spectroscopy for Organic compounds (II M. Sc), Inorganic Chemistry Laboratory (II M. Sc)

Research guidance:

Ph.D: 13 Completed, 7 ongoing M.Sc:3, Post doc 3 (completed), 2 (ongoing)

Completed: M.Sc: 17 and Int. MSc: 11

(Details are given in Annexure – B)

Membership of professional Bodies:

Luminescence Society of India (LSI) – Life member

Chemical Research Society of India (CRSI/LM 2851) – Life Member

Materials Research Society of India (MRSI/LMB3398) – Life Member

Invited Seminars/Talks

(Details are given in Annexure – A)

Sponsored/ consultancy Projects (completed/ongoing/submitted):

S. No	Title of the Project	Sponsoring agency	Amount	Role (PI/Co-PI)	Status
1.	Rational design and synthesis of optoelectronic Lanthanide based molecular materials for OLEDs	BRNS, DAE (Sep 2012 – Sep 2015)	17 Lakhs	PI	Completed
2.	Molecular designing and synthesis of highly efficient red emitting Iridium based dendrimers for OLED applications	DST (Jan 2013 – Dec 2015)	24.9 Lakhs	PI	Completed
3.	Lanthanide-based molecular complexes and their applications	TEQIP II – travel support to visit the Department of Chemistry, Wayne State University,	4.2 Lakhs	(PI and graduate student)	Completed
4.	Rational design and synthesis of Iridium based molecular materials for Organic Electronics	INSPIRE, DST (Jan 2013 – Dec 2017)	35 Lakhs	PI	Completed

5.	Lanthanide-based wavelength converters for smart lightings	NRB, DRDO and NIT Rourkela (Jan 2015 – Dec 2018)	43.5 (25.76 + 17.74) Lakhs	PI	Completed
6.	Metal-free deep blue emitting fluorophores for organic light-emitting diodes: approaching NTSC standard	EMR – SERB, DST, India	42.29 L	PI	Completed
7.	Molecular Engineering of Organic Antennas and their Impact on Dual Emissive Lanthanide Complexes for Smart White LEDs and Molecular Thermometers	CRG-SERB-DST	46.46 L	PI	Completed
8.	Color-Tunable Thermally Activated Delayed Fluorescent Emitter for OLED Applications	IIT Hyderabad	30 L	PI	Ongoing
9.	Rational Design and Synthesis of Ultrabroad Band Near-Infrared Emitting Phosphors for Light-Emitting Diodes	BRNS	28,29,500	PI	Ongoing
10.	Molecular Design and Synthesis of Dual-Emissive Lanthanide Complexes for White Light Emitting Diodes and Luminescent Thermometer	DST (WISE)	35,69,400	PI	Ongoing
11.	Red to NIR Emissive Lanthanide Complex Liquids at Room-Temperature for Soft Displays and Biomedical Applications	IITH-NIMS	Funding for a Graduate student exchange (~1 year) and a PI visit as a guest researcher (1 month)	With Dr. Takashi Nakanishi, Group Leader at Research Center for Materials Nanoarchitectonics	Ongoing
12.	Rational Molecular Engineering of AIE-Driven HLCT Blue and TADF-Integrated Lanthanide-Based Red-NIR Emitters with Enhanced Color Purity for Optoelectronics	DST-ANRF-ARG	237 L	PI	Under review
13.	Integrating Computational Modelling and Experimental Verification: Rational Design and Development of AIEgens-based Multimodal Organic Phototheranostic Applications	DST-ANRF-ARG	~49 L	Co-PI	Under review
14.	Molecular Engineering of Heavy-Metal-free Red-NIR TADF Organic Emitters for Advanced Optoelectronic Applications	CSIR-EMR-II	~47 L	PI	To be submitted (10 th Nov25)
15.	Lotus Proposal submitted Aug 2025	Prof. NAKANISHI Takashi, NIMS, Japan Prof. C. Adachi, Kyushu University, Japan			

Short-term course/Conference organized:

S. No	Title	Place & Dates	No. of Participants	Role
1.	Hosted National Integration Camp (NCC cadets)	NIT Rourkela, Dec 19 – 30, 2013.	(600 cadets from all over India)	Coordinator
2.	DST sponsored INSPIRE Science camp	NIT Rourkela, Nov 30 – Dec 5, 2013.	400	Coordinator
3.	DST sponsored INSPIRE Science camp	NIT Rourkela, Dec 6 - 11, 2014.	250	Coordinator
4.	Hosted National Integration Camp (NCC cadets)	NIT Rourkela, Dec, 2014.	(600 cadets from all over India)	Coordinator
5.	DST sponsored INSPIRE Science camp	NIT Rourkela, Dec 6 - 11, 2015.	300	Coordinator
6.	DST sponsored INSPIRE Science camp	NIT Rourkela, May 8 - 12, 2017.	350	Coordinator
7.	Webinar on Frontiers in Organic electronics	NIT Rourkela, 29th June to 3rd July 2020	850	Coordinator
8.	Webinar on Frontiers in Chemical Sciences	NIT Rourkela, 6th to 10th July 2020	850	Coordinator
9.	International Webinar on Frontiers in Organic electronics	NIT Rourkela & IIT Roorkee, 1 st April – 2 nd April 2022	300	Convener
10.	Workshop on Smart Functional Materials 2024	17 th Aug 2024, IIT Hyderabad	100	Convener
11.	International Conference on Main-group Molecules to Materials (MMM III)	9-11, Dec 2023	~300	Co-Convenor

Projects: (Active member of the team of researchers in the successful completion of the following projects)

1. Synthesis of new lanthanide-containing Metallo polymers as emitters and charge transporters in Organic and Polymer Light-Emitting Diodes – supported by **CEA through nanoscience program – April 2009 – March 2010**
2. Research of its Core Materials and Processing for Low-cost, Flexible Optoelectronic Signage - supported by the ***Korean government through the World-class University program – Dec 2008 – Aug 2013***
3. Designing new high-performance phosphor materials for energy-efficient Solid State Lighting – supported by the ***Korean government through the EEWS program – Jan 2008 – Dec 2008***

4. Development of highly efficient blue phosphor for CCFL applications – supported by **CMS Company – April 2007 – March 2009**
5. Development of high efficiency and durability new blue emitting phosphors for PDP application and improvement of luminescent properties of new blue emitting phosphors – supported by **LG electronics – April 2007 – March 2008**
6. Environmentally Secure Rare Earth-Based Colorants for Surface Coating Applications sponsored by NMTLI – CSIR, New Delhi (Council of Scientific Industrial Research) in the Indian Institute of Technology, Madras, Chennai, India. (Phase I) – **Aug 2003 – Dec 2005**
7. Direct Methanol Fuel Cell sponsored by **MNES, New Delhi** (Ministry of Nonconventional Energy Sources) in Spic Science Foundation, Chennai, India – **Oct 2002 – May 2003**
8. Regenerative Fuel Cell sponsored by **ISRO, Bangalore** (Indian Space Research Organization) in Spic Science Foundation, Chennai, India – **Oct 2002 – May 2003.**

Potential Reviewer for the journal: Journal of Physical Chemistry C, Inorganic Chemistry, Dalton Transactions, Physical Chemistry Chemical Physics, RSC Advances, Journal of Materials Chemistry C, Inorganic/Materials Chemistry Frontiers, Journal of Alloys and Compounds, Journal of Luminescence, Materials Letters, Scientific reports, Dyes and pigments, Organic electronics, Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy, Chemistry select, Asian Chemistry Journal, Luminescence, Journal of Materials Science, Journal of organic chemistry. Advanced functional materials, Advanced optical materials, Advanced Materials, Angewandte chemie International Edition.

Personal details:

Date of birth	: 05 – 06 - 1980
Age	: 45 years 2 months
Nationality	: Indian
Father's name	: (Late) M. Vaidyanathan
Marital status	: Married
Blood Group	: B Positive
Passport number	: C3078500 (OLD PP No. F2618175, Z3108870)

Annexure "A"

Invited Seminars/Talks

Invited Talk, Molecular Engineering of Inorganic-Organic Hybrid Luminogens for Light emitting Diodes, Department of Engineering Science, The University of Electro-Communications, 1-5-1 Chofugaoka, Chofu 182-8585 Japan, July 2025

Invited Talk, Molecular Design and Synthesis of Dark Triplet Exciton Harvesters for Optoelectronics, Nihon University, Japan, June 2025

Invited Talk, Proactive design and synthesis of Luminogens for Smart Lightings, NIMS, Japan, June 2025

Invited Talk, Proactive design and synthesis of Luminogens and their versatile applications, April 2025, IISER, Tirupati.

Invited Talk, Proactive design and synthesis of Luminogens and their versatile applications, ICSTAR, 2025, SV University, Tirupati.

Invited Talk, Molecular Siblings: Single Component White Light Emissive Molecular Europium Complexes for Light Emitting Diodes, Feb, 2025, The 2025 Pure and Applied Chemistry International Conference (PACCON202), Thailand.

Invited Talk, Proactive design and synthesis of Luminogens for Smart Lightings, Vidyasirimedhi Institute of Science and Technology (VISTEC). Wangchan Valley 555 Moo 1 Payupnai, Wangchan, Rayong 21210 Thailand

Invited Talk, Molecular Siblings: Single Component White Light Emissive Molecular Europium Complexes for Light Emitting Diodes, The Pure and Applied Chemistry International Conference 2025 Chemistry for a Changing World, Feb 2025

Invited Talk – "Molecular Design and Synthesis of Triplet Exciton Harvesters for Efficient Organic Light-Emitting Diodes" – Tokyo Institute of Technology, Frontier Materials Research Center, Majima & Izawa Laboratory, 4259 Nagatsuta-cho, Midori-ku, Yokohama, Kanagawa 226-8503, Japan – 85th Oct 2024

Invited Talk – "Proactive design and synthesis of Luminescent Materials for smart lighting" – Center for Organic Photonics and Electronics Research (OPERA), Faculty of Engineering, Kyushu University, Fukuoka, Japan – 25th Oct 2024

Invited Talk – "Luminescent Materials and their applications" – Department of Chemistry, Thiruvalluvar University, Vellore – Tamil Nadu – Sep 2024

Invited talk – Motivational talk "Pursuing higher education" – Department of Chemistry, MGAC College – Vellore, Tamil Nadu – Sep 2024

Invited talk - "Luminescent Materials and their Applications" – Department of Chemistry, VIT-AP University, AP – August 2024 – Webinar on Advanced Materials 2024.

Invited Talk – "Functional Materials for Organic Light Emitting Diodes (OLEDs)", Refresher Course – University of Hyderabad – June 2024

Invited Talk – "Luminescent materials: Fundamentals to applications", Refresher Course – University of Hyderabad – June 2024

Invited talk – "Luminescent Material for solid-state lightings," NIT Jaipur, March 2024.

Invited talk – “Molecular Design and synthesis of new TADF emitters for highly efficient solution-processed white Organic Light Emitting Diodes”, **CRSI Meeting at MKU University, Tamil Nadu, Feb 2024.**

Invited talk – “Luminescent Materials: Fundamental to applications”, **Faculty development programme, Hyderabad Central University, Jan 2024.**

Invited talk – “Lanthanide-based Luminescent Materials for solid-state lighting”, MTIC 2023, IISc Bangalore.

Invited talk - “Dual emissive Lanthanide complexes through Excited state modulation of organic antennas for smart lightings”, 3rd OCS-ONEDAY EXTENDED LECTURE SERIES 2023" on the theme "Research in Frontier Areas of Chemical Science and Technology" KIIT University, 5th Dec - 2023

Invited talk - "Excited state modulation of organic antennas and their impact on dual emissive Lanthanide complexes for solid-state lightings" VIT Vellore, School of Chemical Sciences, Faculty development Programme – Oct 17, 2023

Invited talk - "**Luminescence materials: Fundamental to applications** in Refresher course in Physics and , Organized by UGC-Human Resource Development Center, Sri Venkateswara University, Tirupati – 13th Oct 2023

Invited talk - "**Lanthanide based Luminescence materials and their applications**" in "**Chemistry Association Guest Lecture**" series at Department of Chemistry, SRM University – Oct 10, 2023

Invited talk - "**Lanthanide based Luminescence materials for smart lightings**" VIT Vellore, School of Chemical Sciences, Faculty Development Programme – Aug 4, 2023

Invited talk - "**Molecular Design and synthesis of new TADF emitters for highly efficient solution-processed TADF OLEDs**" IISER Mohali, School of Chemical Sciences – June 13, 2023

Invited talk - "**Lanthanide based Luminescence Materials and their Applications**" IIT Mandi, School of Chemical Sciences – June 13, 2023

Invited talk - "**Molecular Design and synthesis of new TADF emitters for highly efficient solution-processed TADF OLEDs**" International Conference on Functional Materials for Future Technology, (IC-FMFT 2023) organized by Centre for functional Materials, VIT University – 19th – 21st April 2023

Invited talk - "**Lanthanide based Luminescence materials and their Applications**" National Conference on Signal Processing, Sustainable Energy Materials and Astronomy And Astrophysics (NSSEMA - 2023) is organized by School of Studies in Physics & Astrophysics and School of Studies in Electronics & Photonics, Pt. Ravishankar Shukla University, Raipur, C.G. India during March 16-18, 2023

Invited talk - "**Luminescence materials and their Applications**" 28/02/2023 **STC** on "Optical Materials and Characterization Techniques & Synchrotron Radiation and Neutron based Characterization Techniques" during 24/02/2023 to 03/03/2023 organized by Department of Physics, National Institute of Technology, Uttarakhand.

Invited talks (rising star) – Novel Acceptor Interlocked Molecular Design for highly efficient solution-processed TADF OLEDs – The 7th International workshop on TADF, Japan, Dec 1-2, 2022.

Invited talks – Lanthanide based phosphors for smart lightings, Centre for Functional Materials, VIT Vellore. 31st October 2022 (Virtual Seminar Series on Functional Materials 2022).

Invited talks – “Luminescent materials and their applications in light emitting diodes (LEDs)” Refresher Course in Physics organized by Human Resource Development Centre of Bharathiar University, 11-13, Oct 2022

Invited talks – “Functional Materials for light emitting diodes (LEDs)” National Level webinar on Frontiers in functional materials organized by Marudhar Kesari Jain College for Women, Vaniyambadi - 635 751, Tamil Nadu, INDIA

Invited talks – 1st International webinar on organic electronics – Aug 24-26, 2022 organized by Sharada University, New Delhi.

Invited talks – “Functional Luminescent Materials and their applications in display and lighting technology”- Webinar on “Functional Materials and Nanostructures” organized by the Department of physics, Islamiah College (Autonomous), Vaniyambadi – 22nd December 2021

Invited talks – “Functional Materials and their applications in light emitting diodes”- FDP on Emerging trends in chemical sciences, organized by Saranathan College of Engineering, Trichy, India – 23-28 Aug 2021

Invited talks – “Functional Materials for Blue or white OLEDs applications”- Departmental invited talk, organized by VIT University, Vellore, India – July 2021

Invited talks – “Luminescent Materials for lighting applications”- Departmental invited talk, organized by Thiagarajar College of engineering, Madurai, India – 21st April 2021

Invited talks – “Luminescent Materials for lighting applications”- International Webinar on Advanced Functional Materials (18th-21st August 2020) - Organized by Department of Chemistry & Physics Nizam College, Osmania University, Hyderabad-500001

Invited talks – “Phosphor Materials for lighting applications”- one day national workshop on Recent trends in engineering materials RTEM 2020 – 7TH Aug 2020.

Invited talks – Molecular Materials and their applications in light emitting diodes - “Molecular Materials: Present and Future Trends” – virtual symposium organized by Annamalai University, Tamil Nadu, India – 15th July 2020.

Invited talks – Molecular Materials and their applications in light emitting diodes - International Webinar Series on “Futuristic Medicinal and Materials Chemistry –FMMC-2020” (June 2020) organized by Department of Chemistry, B.S.Abdur Rahman Crescent Institute of Science and Technology, Chennai-600 048, Tamil Nadu, India.

Keynote Invited talks – “Design strategy of organic blue fluorophores for Organic Light Emitting Diodes”- 2nd International Conference on Recent Trends in Renewable Energy & Sustainable Development (ICRTRESD-2020) - 30th & 31st January 2020, BIT Raipur.

Invited talks – “Organic fluorophores for efficient deep-blue and white Organic light emitting diodes”- Organic molecules as synthons & reagents for innovations (OMSRI-2019), Feb 8 – 9, 2019, IIT Roorkee.

Invited talks – “Blue organic light emitting diodes: Challenges and future perspectives”- International conference on luminescence and its applications (ICLA 2019), Pt. Ravishankar University, Raipur, Jan 2018

Invited talks – “Rational design and synthesis of organic fluorophores for blue organic light emitting diodes”- BARC, Mumbai, Dec 2018

Invited talks – “Organic luminophores for blue OLEDs”- VIT University, Vellore, Nov 2018

Invited talks – “Eu^{III} Complexes for White Light Emitting Diodes and Ratiometric temperature sensors”- National Conference on Novel Materials for Device Applications (NCNMDA-2018) from November 4-5, 2018 to be held at S. V. University, Tirupati, A.P, India.

Invited talks – “Luminescent europium complexes for light emitting diodes”- BARC, Mumbai, April 2018

Invited talks – “Functional molecular materials and their versatile applications”- School of Advanced Sciences, Department of Chemistry, VIT University, Vellore – Sep 2017

Invited talks - Short term course on Waste Heat Recovery and Thermal Energy Storage (WHRTES-2017) “Functional Materials for Energy applications” at Department of Mechanical Engineering, NIT Rourkela, India

Invited talks - Department of Chemistry Pondicherry University, Pondicherry, Feb-2017, “Inorganic luminescent materials for energy efficient solid state lighting (SSL)”, at Center for Green Energy Technology”

Invited talks - Madanjeet School of Green Energy Technologies, Pondicherry University, Feb-2017, “Design strategy to obtain highly luminescent molecular materials for lighting applications”

Invited talks - National Conference on Advances in Chemical Sciences and Technology (NCACST-17), Feb 2-3, 2017, KCG College of technology, Chennai, “Design strategy to obtain highly luminescent molecular materials for lighting applications”

Invited talks – Sri Venkateshwara University, Tirupati, August 2016 “Carrier Opportunity and Higher Education Guidance – In Basic Sciences”

Invited talks – Vellore Institute of Technology (VIT) University, Vellore, May 2016 “Controlled energy transfer from ligand to Eu (III) ion: A unique strategy to obtain bright white light emitting Eu-complex”

Invited talks – NIT, Raipur, April, 2016 “Proactive design of host materials for phosphorescent OLEDs: An overview”

Invited talks – NIT, Raipur, April, 2016 “Light emitting lanthanide Complexes, fundamental to applications: A Bird's Eye View”

Invited talks – National conference on Luminescence and its applications (NCLA 2016), Nagpur, Feb 2016 “Proactive design of host materials for phosphorescent OLEDs: An overview”

Invited talks – National Institute of Technology, Rourkela, Dec 2016 “Carrier Opportunity and Higher Education Guidance – In Basic Sciences”

Invited talk – Department of chemistry, Wayne state University, Detroit, USA, Feb 2015 “Molecular materials for lighting applications”.

Invited talks – National Institute of Technology, Rourkela, Dec 2014 “Carrier Opportunity and Higher Education Guidance – In Basic Sciences”

Invited talks – National Institute of Technology, Rourkela, Dec 2013 “Carrier Opportunity and Higher Education Guidance – In Basic Sciences”

Invited talk – National conference on Luminescence and its applications (NCLA 2013), PSU college of engineering, Bangalore, Feb 2013 “Iridium molecular complexes for OLED applications”

Invited Technical talk – International conference on Luminescence and its applications (ICLA 2012), ICT & RGTU, Hyderabad, Feb 2012 “Eu³⁺ substituted double perovskites: Novel orange-red emitting phosphors for Solid state White LEDs”

Invited talk – FDP on Catalysis: Concepts to Applications, NIT, Calicut, June 2 – 6, 2012 “Bio-Inspired Photocatalysts for Hydrogen Production”

Invited talks – FDP on Display materials and its applications, organized by RNS Institute of Technology, Bangalore, April 12 – 13, 2012 “Lamp phosphors: concepts to applications”

Invited talk – UGC sponsored National seminar on advanced materials (NASAM 2012) at Isalmiah College, Vaniyambadi – 4 – 5 March 2012 “Lanthanide-based materials for solid state lighting”

Invited talks – GTER 2012 at Gurukula Kangri Vishwavidyalaya, Haridwar – 11 – 13 Feb 2012 “Europium activated oxides - Alternative Red phosphors for energy efficient solid state lighting”

Invited talks - FDP on Materials Present and Future perspectives at NIT Calicut – 12 to 16 Dec 2011 “Lanthanide activated luminescent materials as wavelength convertors – lighting and display applications”

Invited talks - FDP on Current trends in Organometallic Chemistry at NIT Calicut – 4 to 8 June 2011, “Lanthanide containing luminescent organometallic complexes and its specific applications” and “Luminescent organometallic complexes for organic light emitting diodes”

Annexure “B”

List of Publications (in reverse chronological order)

Patent Filed:

1. **Sivakumar Vaidyanathan**, Jaipal G, S R Nayak, Sabita Patel, “Near UV-deep blue emitters and uses thereof in organic electronics,” Indian patent filed.
2. **Sivakumar Vaidyanathan**, Md Intekhab Alam, Sabita Patel, “Yellow TADF emitters and uses thereof in organic electronics”, Indian patent filed.
3. **Sivakumar Vaidyanathan**, Maredi Swetha, “Solid-state single-component white light-emissive lanthanide complexes and their usage in white LEDs”, Submitted

4. **Sivakumar Vaidyanathan**, Diksha Thakur, Biswaranjan Sahu, "Chiral TADF Emitters with Enhanced Luminescence Dissymmetry Factor for Circularly Polarized Luminescence-based devices", Submitted
5. **Sivakumar Vaidyanathan**, Vara Prasad Rebaka, Samatha Devineni, "Single-Component Sunlight-Mimicking White Light Emissive Lanthanide Complexes: Synthesis and Applications Thereof", Submitted

Book chapters:

1. **Sivakumar Vaidyanathan**, P Pradhan, Lanthanide-Based Molecular Thermometers: An Overview, Luminescence: Theory and Applications of Rare Earth Activated Phosphors, 45, 2021, <https://doi.org/10.1515/9783110676457-002>, © 2021 Walter de Gruyter GmbH, Berlin/Boston
2. **Vaidyanathan Sivakumar**, R Marikumar, S Kasturi, S Kumar, White Light Emitting Phosphors for Solid State Lighting, Emerging Synthesis Techniques for Luminescent Materials, 150-250, 2018, DOI: 10.4018/978-1-5225-5170-6.ch006, IGI Global
3. **Sivakumar Vaidyanathan**, J Tagare, Phenanthroimidazole-Based Organic Fluorophores for Organic Light-Emitting Diodes: An Overview, Emerging Synthesis Techniques for Luminescent Materials, 305-388, 2018, DOI: 10.4018/978-1-5225-5170-6.ch009, IGI Global
4. **Vaidyanathan Sivakumar**, B Rajamouli, Molecular Designing of Luminescent Europium-Metal Complexes for OLEDs: An Overview, Phosphors, 325-404, 2018, Jenny Stanford Publishing
5. **Vaidyanathan Sivakumar**, S Kasturi, Eu³⁺-Based Orange-Red-Emitting Inorganic Color Converters: An Overview, Phosphors. 1, 219-324, 2018, Jenny Stanford Publishing

Submitted papers:

1. Md Intekhab Alam, **Sivakumar Vaidyanathan**, "Dual Acceptor based Molecular design for TADF covering the whole visible spectrum and study of their excited state photophysical phenomenon: A theoretical approach"- Submitted.
2. Sibani Mund and **Sivakumar Vaidyanathan***, Recent Advances and Future Perspectives in Efficient Light Harvesting Organic Antennas based Europium(III) Complexes: Serving as a Boon for Next Generation Technology, Submitted
3. Diksha Thakura, Biswaranjan Saha, Prathap R, Pradyumna Sahoo, Mahesh Kumar Ravva, Sabita Patel, and **Sivakumar Vaidyanathan***, Molecular Design and Synthetic Pathways for Chiral/Achiral TADF and HLCT Emitters in Near Infrared Region: Recent Progress and Future Perspective, Submitted
4. Bhabana Priyadarshini Debata, Atthapon Saenubol, Vinich Promarak, Sabita Patel, and **Sivakumar Vaidyanathan***, Molecular Engineering of Benzimidazole and Phenanthroimidazole Donor- π -Acceptor Fluorophores with Tunable HLCT-AIE Emission for Multifunctional Applications, Submitted

5. Bhabana Priyadarshini Debataa, Sabita Patel, and **Sivakumar Vaidyanathan***, Fluorophores with Imidazole Building Blocks and Their Multifunctional Platforms for Optoelectronics, Chemo-Sensing, and Security Applications: Recent Advances and Future Perspectives, **Submitted**
6. Diksha Thakur, Biswaranjan Sahu, Wijk Yospanya, Reiko Oda, **Sivakumar Vaidyanathana***, Linkage Isomers and Their Enantiomeric Siblings as CPL-TADF Emitters: Modulating Chirality and Enhancing Dissymmetry Factor Through N- and O-Linked Carbazoles, **Revision to be submitted**
7. Swetha Maredi, Rohit B, Takashi Nakanishi, and **Sivakumar Vaidyanathana***, Hetero Atom Replaced by Phenyl Insertion in Molecular Eu-complex Facilitated Sun Light-like Emission for Sustainable White LEDs and Non-Contact Thermometer, **Under Review in ICF.**
8. Bhabana Debata, Khitisruta Nayak, Manjit Dehury, and Sabita Patel, **Sivakumar Vaidyanathana***, HLCT/HLCT-AIE-Enabled Donor-Acceptor System through Locking and Unlocking Strategy and Its Versatile Applications, **Revision to be submitted in JMCC**
9. Vara Prasad Rebaka, Samatha Devineni, Swetha Maredi, Sandhya Rani Nayak, **Sivakumar Vaidyanathana***, Engineering Ligand Spatial Arrangement Enabled Broad-Spectrum White Light Emission mimicking sunlight from Donor-Acceptor Europium(III) Complexes, **Revision to be submitted in JMCC**
10. Priyadarshini Pradhan and **Sivakumar Vaidyanathan***, Luminescent Eu(III) Molecular Complexes: Synthesis, Characterization, Photophysical, Theoretical Analysis, and their Versatile Applications, **Submitted.**
11. Md Intekhab Alam, Vinod Kumar, Ajay Perumal, **Sivakumar Vaidyanathan**, "Enhancing ultra-long pure organic room temperature phosphorescence by simple aromatic carbonyl: combined experimental and theoretical investigations"- **under revision.**
12. A. B. Kajjam, K. Singh and **V. Sivakumar***, Excitation-dependent organic down-converting Imidazo-pyridine based materials for WLEDs: Combined experimental and theoretical investigation, - *Journal of Information Display*, **under revision.**
13. Rajamouli Boddula, Kasturi Singh, and **Sivakumar Vaidyanathan***, Designing organic chromophores and studying their excited photophysical properties for Eu^{III} complexes – a theoretical approach, submitted.
14. K. Singh, R. Boddula, A. B Kajjam, M. J. Allen, **Sivakumar Vaidyanathan*** Effect of N-Conjugation in Imidazo-Bipyridyl Ancillary Ligands and Their Energy-Transfer Impact on Eu(III)–Beta-Diketonate Complexes for White-Light-Emitting Diodes. Submitted.

Google Scholar Link:

https://scholar.google.co.in/citations?hl=en&user=LSrFzcAAAAJ&view_op=list_works&sortby=pubdate

Total = 125 (peer review journals), Citations – 3975, h-index – 38, i10-index – 86

In Refereed International Journals:

1. Kesavan, H. Lee, Shahnawaz, S.Swyamprabha, D. Tavgeniene, D. Blazelevicius, G. Krucite, J. Jayachandran, S.R.Nayak, **Sivakumar Vaidyanathan**, S. Grigalevicius, Juo-Huei Jou, "Solution Processable Fluorene-Based Hole Transporting Materials for High-Efficiency OLEDs" *Synthetic Metals*, 314, 2025, 117921, (Citation – 0), Impact factor: 4.6 **Q1**
2. S Mund, K Singh, R. Devi, U. Subuddhi, **Sivakumar Vaidyanathan***, Modulating Triplet States in Light-Harvesting Antennas: Multicolor (Including White) Emission from Molecular Europium Complex Siblings, *Chemistry - An Asian Journal*, 2025, e00830, <https://doi.org/10.1002/asia.202500830> (Citation – 0), Impact factor: 3.3 **Q2**
3. P Sharma, JP Madda, **Sivakumar Vaidyanathan***, Synthesis and optical properties of LiCaY (MoO₄)₃:Eu³⁺ red phosphor with negligible concentration quenching for highly efficient white LED and plant growth applications, *Dalton Trans.*, 2025, **54**, 14868-14881 DOI: [10.1039/D5DT01050K](https://doi.org/10.1039/D5DT01050K), (Citation – 0), Impact factor: 3.3 **Q1**
4. S Maredi, SR Nayak, MI Alam, D Thakur, **Sivakumar Vaidyanathan***, "Broad-band Emissive Phenanthroimidazole-based Donor-Acceptor Luminogens for Hybrid White Light Emitting Diodes and Sensors for Picric acid detection", *J. Mater. Chem. C*, 2025, **13**, 14385 – 14403, DOI: [10.1039/D5TC00780A](https://doi.org/10.1039/D5TC00780A) (Citation – 1), Impact factor: 5.1 **Q1**
5. R. Devi, K Singh, S Mund, S Devineni, **Sivakumar Vaidyanathan***, 'Efficient Narrow Band Pure Red Emitting Molecular Europium Complexes for White Light-Emitting Diodes and Anticounterfeiting Applications', *J. Mater. Chem. C*, 2025, **13**, 15519-15531, DOI: [10.1039/D5TC00861A](https://doi.org/10.1039/D5TC00861A), (Citation – 0), Impact factor: 5.1 **Q1**
6. Bhabana Debata, J Dash, Sabita Patel, and **Sivakumar Vaidyanathan***, "HLCT-AIE active Deep Blue Fluorophores and their versatile applications: A Multifunctional Approach for Advanced White LED Materials, Picric Acid Sensing and Fingerprint Visualization", *J. Mater. Chem. C*, 2025, **13**, 14385-14403, DOI: [10.1039/D5TC01672J](https://doi.org/10.1039/D5TC01672J). (Citation – 2), Impact factor: 5.1 **Q1**
7. Snigdhamayee Rana, Sandhya Rani Nayak, Atthapon Saenubol, Vinich Promarak, Sabita Patel, **Sivakumar Vaidyanathan*** "Synergistic enhancement of Non-doped OLED efficiency with EQE ~ 15% in donor- π -acceptor based blue emitters featuring combined AIE-HLCT characteristics", *Advanced Functional Materials*, 2025, 35(42), 2507011, <https://doi.org/10.1002/adfm.202507011>, (Citation – 7), Impact factor: 18.5 **Q1**
8. Diksha Thakur, **Sivakumar Vaidyanathan***, Chiral Lanthanide Complexes in the History of Circularly Polarized Luminescence: A Brief Summary, *J. Mater. Chem. C*, 2025, **13**, 9410 – 9452, <https://doi.org/10.1039/D5TC00097A> (Review Article), From the themed collection: *Journal of Materials Chemistry C* Recent Review Articles (Citation – 6), Impact factor: 5.700 **Q1**

9. Bhabana Priyadarshini Debata, S. Patel and **Sivakumar Vaidyanathan***, Imidazo[1,5-a]pyridine-benzilimidazole conjugated (donor- π -acceptor) greenish-yellow fluorophores and their applications in white light-emitting diodes, acidochromism and anticounterfeiting, *J. Mater. Chem. C*, 2025, **13**, 10413-10433 DOI: 10.1039/D5TC01152C, (Citation – 3), Impact factor: 5.700 **Q1**
10. Swetha Maredi, Diksha Thakur, Dhruva Jyoti Boruah, Sandhya Rani Nayak, Panneerselvam Yuvaraj, and **Sivakumar Vaidyanathan***, "HLCT Driven Twisted Benzo[d]thiazole Based D–A–D Fluorophores for High-Performance, Two-Component White LEDs" *Chem Asian Journal*, 2025, **20**(12), e202500152 (Citation – 0), Impact factor: 3.3 **Q1**
11. Amit Kumar Mahto, Joyoti Ghosh, Priyadarshini Pradhan, Moloy Sarkar, **Sivakumar Vaidyanathan**, Rajeev Ramanan, Jaya Prakash Madda, Fluorescence Enhancement in a Tri(biphenyl-4-yl)amine Derivative via Solid-State Grinding with Metal Salts: Chemosensing and White Light-Emitting Diode Applications, *ChemPhotoChem*, 2025, **9**(6), e202500021, <https://doi.org/10.1002/cptc.202500021> (Citation – 1), Impact factor: 3.00 **Q1**
12. Priyadarshini Pradhan and **Sivakumar Vaidyanathan***, "Highly efficient red/orange-red emitting Eu^{3+} and $\text{Sm}^{3+}/\text{Eu}^{3+}$ co-doped phosphors with their versatile applications" - *Dalton Trans.*, 2025, **54**, 6060-6080, DOI: 10.1039/D5DT00004A, (appeared in back cover page)(Citation – 6), Impact factor: 4.569 **Q1**
13. Bhabana Priyadarshini Debata, M. D. Thiagarajan, Prathap R, J. D. Girase, S. Kulathu Iyer, S. Patel and **Sivakumar Vaidyanathan***, Benzil-imidazole Blue Fluorophores and their applications in Blue/White Light-Emitting Diodes, Sensing and Anticounterfeiting, *J. Mater. Chem. C*, 2025, **13**, 2711 – 2731, <https://doi.org/10.1039/D4TC04661G> (Citation – 6), Impact factor: 5.700 **Q1**
14. Sibani Mund, K. Singh and **Sivakumar Vaidyanathan***, "High-performance white light-emitting diodes based on Efficient Trivalent Europium molecular complex", *J. Mater. Chem. C*, 2024, **12**, 18401-18415, (Citation – 4), Impact factor: 5.700 **Q1**
15. Md Inekab Alam, Mangey Ram Nagar, Jwo-Huei Jou and **Sivakumar Vaidyanathan***, "Twisted Acceptor Core Molecular Design with Phenoxazine and Phenothiazine Donors Enabled Yellow TADF Emitter/Sensitizers for Long-lifetime Solution-processed OLEDs Exceeding a 31% EQE". *Chem. Mater.* 2024, **36**, 18, 8649–8662, Dedicated to Professor. U. V. Varadaraju on the occasion of her 70th Birthday, (Invited article, part of The Chemical Research Society of India (CRSI) celebrated its 25th anniversary in 2024), (Citation – 7), Impact factor: 7.2 **Q1**.
16. S. Mund, K. Singh, Deepak, Anju Babu, U. Subuddhi and **Sivakumar Vaidyanathan***, "Unveiling the versatility of Coumarin Integrated with Phenanthroline/Thiabendazole based Eu^{III} complexes for smart LEDs, Vapoluminescence, and Bioimaging applications", - *ACS Appl. Bio Mater.* 2024, **7**, 9, 5795–5809, DOI: [10.1021/acsabm.4c00839](https://doi.org/10.1021/acsabm.4c00839), (Invited article, part of The Chemical Research Society of India (CRSI) celebrated its 25th anniversary in 2024), Dedicated

to Professor A. K. Mishra on the occasion of her 65th Birthday, (Citation – 3), Impact factor: 4.7, Q1

17. S. Rana, **Sivakumar Vaidyanathan*** and S. Patel, Aggregation Induced Emission (AIE) based Donor- π -acceptor Fluorophores: An approach to fabricate acidochromic sensors and white light-emitting diodes, DOI: 10.1039/D4TC02617A, *J. Mater. Chem. C*, 2024, **12**, 14148-14164, (Citation – 8), Impact factor: 5.700 Q1
18. Diksha Thakur, **Sivakumar Vaidyanathan***, Chirality Inducing Units in Organic TADF Molecules: A Way to Circularly Polarized Luminescence, *J. Mater. Chem. C*, 2024, **12**, 13168-13229, DOI: 10.1039/D4TC01923G (Review Article), (Citation – 5), Impact factor: 5.700 Q1
19. Alam, Md Intekhab; Nagar, Mangey Ram; D. Barman, P. K. Iyer, Jou, Jwo-Huei; **Vaidyanathan, Sivakumar***, "Tailoring the structural rigidity utilizing lock/unlock donor strategy for highly efficient Solution processed Blue and Green HLCT OLEDs" – *J. Mater. Chem. C*, 2024, **12**, 13585-13595, <https://doi.org/10.1039/D4TC00417E>, (Citation – 11), Impact factor: 5.700 Q1
20. Swetha Maredi, Sibani Mund, Sandhya Rani Nayak, Samatha Devineni, Subrahmanyam C, **Vaidyanathan Sivakumar***, White Light Emissive Eu(III) Complexes through Ligand Engineering and their Applications in Cool Near Ultraviolet White Light Emitting Diodes and Thermometer, *Chemphyschem*, 2024, 25(16), e202400320, doi: 10.1002/cphc.202400320, Invited article. (Citation – 7), Impact factor: 2.2 Q2, Dedicated to Professor Archita Patnaik on the occasion of her 65th Birthday
21. Bhabana Priyadarshini Debata, Jwo-Huei Jou, Sabita Patel and **Sivakumar Vaidyanathan***, The design and synthesis of 1-phenylimidazo[1,5-a]pyridine – anthracene based fluorophore for greenish-yellow organic light emitting diode and warm white LED, - *Journal of Information Display*, **2024**, 25(4), 319-331 DOI:10.1080/15980316.2024.2304869 (Citation – 2), Impact factor: 3.4 Q1
22. Snigdhamayee Rana, Sandhya Rani Nayak, Sabita Patel and **Sivakumar Vaidyanathan***, "Efficiency Boost in Non-Doped Blue Organic Light Emitting Diodes: Harnessing Aggregation-Induced Emission - A Comprehensive Review" - *J. Mater. Chem. C*, **2024**, **12**, 765-818, DOI: 10.1039/D3TC03449F, (Citation – 24), Impact factor: 5.1, Q1
23. Sandeep Kumawat, Sunidhi Singh, Tharun Bhatt, Anjali Maurya, **Sivakumar Vaidyanathan**, Kishore Natte, and Jagadeesh Rajenahally, Valorization of Bio-Renewable Glycerol by Catalytic Amination Reactions, (Critical Review) - *Green Chem.*, **2024**, **26**, 3021-3038, DOI: 10.1039/D3GC02699J (Citation – 14), Impact factor: 9.2, Q1
24. Gobinath Marappan, Abdul Kaium Mia, Kishore Puspharaj, **Sivakumar Vaidyanathan**, Yoshiyuki Kawazoe, Yuvaraj Sivalingam, Velappa Jayaraman Surya, "Response to VOCs Stimuli by Triphenylamine Derivatives Functionalized Zinc Oxide Nanorods: A Promising Material for Food Freshness Monitoring" - *Surfaces and*

- Interfaces.*, **2024**, 103648 DOI: doi.org/10.1016/j.surfin.2023.103648, (Citation – 10), Impact factor: 6.4, **Q1**
25. Priyansha Sharma, Jaya Prakash Madda and **Sivakumar Vaidyanathan***, “Narrow band dazzling red emitting (LiCaLa(MoO₄)₃:Eu³⁺) phosphor with scheelite structure for Hybrid White LEDs and LiCaLa(MoO₄)₃:Sm³⁺, Eu³⁺ Based Deep-Red LEDs for Plant Growth Applications” - *Dalton Trans.*, **2023**, 52, 15043-15056, DOI: 10.1039/D3DT02716C, (Citation – 20), Impact factor: 4.569 **Q1**
 26. Kasturi Singh, P. Pradhan, Savita Priya, S. Mund and **Sivakumar Vaidyanathan***, “Recent progress in Trivalent Europium (Eu³⁺) based Inorganic phosphors for solid-state lightings – An Overview” – *Dalton Trans.*, **2023**, 52, 13027-13057, DOI: 10.1039/D3DT00303E (Citation – 44), Impact Factor: 4.569 (Invited Perspective article). **Q1**
 27. Jaipal Devesing Girase, S Singh, BP Debata, SR Nayak, Mangey Ram Nagar, Jwo-Huei Jou, S. Patel and **Sivakumar Vaidyanathan*** “Structural Engineering of Deep-Blue Emitters (Imidazoles Integrated with Triphenylamine) Leads to EQE> 6% and High Color Purity (CIEy~ 0.09) for Solution-Processed OLEDs” - *J. Phys. Chem. C*, **2023**, 127, 33, 16623–16635 DOI: org/10.1021/acs.jpcc.3c03058. (Citation – 9), Impact factor: 3.2 **Q1**
 28. Sandhya Rani Nayak, Jaipal Devesing Girase, Mangey Ram Nagar, Jwo-Huei Jou, **Sivakumar Vaidyanathan***, "Solution Processable Deep-blue OLEDs Based On Benzimidazole-TPA Conjugated Through 9,9-diethyl fluorene (D-π-A) Luminophore With A Hybridized Local And Charge Transfer Excited State", - *J. Phys. Chem. C* **2023**, 127, 21, 10291–10302. DOI: org/10.1021/acs.jpcc.3c00790, (Citation – 19), Impact factor: 3.2 **Q1**
 29. **Sivakumar Vaidyanathan***, “Recent Progress on Lanthanide-based long persistent phosphors: An Overview” – *J. Mater. Chem. C*, **2023**. DOI: 10.1039/D2TC05243A, 11, 8649-8687, This article is part of the themed collection: Journal of Materials Chemistry C Recent Review Articles (Citation – 49), Impact factor: 8.067 **Q1**
 30. Sandhya Rani Nayak, Luke The, Md Intekhab Alam, Mangey Ram Nagar, Jwo-Huei Jou, Sabita Patel, **Sivakumar Vaidyanathan***, "Molecular Modulation by Sigma Conjugated Spacer Enables Efficient Ultraviolet/Deep-blue Emissive Organic Light-Emitting Diodes" - *J. Phys. Chem. C*, **2023**, 127, 7, 3849–3860. DOI: org/10.1021/acs.jpcc.2c08679, (Citation – 6), Impact factor: 3.2 **Q1**
 31. Sandhya Rani Nayak, Sabita Patel, and **Sivakumar Vaidyanathan***, “Imidazole-based fluorescent probes: Concomitant effects of N1 substitution and lone pair on selective recognition of Picric Acid” - *New J. Chem*, **2023**, 47, 3524 - 3534 DOI: 10.1039/D2NJ06079E. (Citation – 11), Impact factor: 2.5 **Q2**
 32. Sandhya Rani Nayak, Shahnawaz, Iram Siddique, Jwo-Huei Jou, sabita Patel, and **Sivakumar Vaidyanathan***, “Multifunctional 4,5-Diphenyl-1H-imidazole-Based Luminogens as Near UV/Deep Blue Emitters/Hosts for Organic Light-Emitting Diodes and Selective Picric Acid Detection”- *J. Phys. Chem. C*, **2023**, 127, 1, 499–515 (Published as part of The Journal of Physical Chemistry virtual special issue

“Early-Career and Emerging Researchers in Physical Chemistry Volume 2) DOI: [org/10.1021/acs.jpcc.2c05220](https://doi.org/10.1021/acs.jpcc.2c05220), (Citation – 6), Impact factor: 4.177 **Q1**

33. Sujith Sudheendran Swyam prabha, Kiran Kishore Kesavan, Iram Siddiqui Dovydas Blazelevicius, Jayakumar Jayachandran, Marius Eidimtas, Sandhya Rani Nayak, Mangey Ram Nagar, Rohit Ashok Kumar Yadav GintareKrucaite, **Sivakumar Vaidyanathan**, Saulius Grigalevicius, Jwo-Huei Jou, “Novel carbazole host materials for solution processed TADF Organic Light Emitting Diodes” - *Dyes and Pigments*, **2022**, 208, 110821, DOI: [org/10.1016/j.dyepig.2022.110821](https://doi.org/10.1016/j.dyepig.2022.110821), (Citation – 8), Impact factor: 4.2 **Q1**
34. Priyadarshini Pradhan, Kasturi Singh, P. Saradhi Maram and **Sivakumar Vaidyanathan***, “Oxide-based red emitting phosphors with high color purity and their versatile applications: synthesis, structure, and luminescence properties” - *ACS Appl. Opt. Mater.*, **2022**, DOI: [org/10.1021/acsao.2c00022](https://doi.org/10.1021/acsao.2c00022) (Citation – 9), Impact factor: N/A, **Q1**
35. Sandhya Rani Nayak, Iram Siddique, Shah Nawaz, Jwo-Huei Jou, and **Sivakumar Vaidyanathan***, “Diphenyl-imidazole based fluorophores for explosive chemosensors and serves as efficient host materials for green Phosphorescent Organic Light Emitting Diodes” - *ACS Appl. Opt. Mater.* **2023**, 1, 1, 132–146, DOI: [org/10.1021/acsao.2c00012](https://doi.org/10.1021/acsao.2c00012). (Citation – 3), Impact factor: N/A, **Q1**
36. Girase, Jaipal Devesing; Kajjam, Aravind; Dubey, Deepak; Kishore Kesavan, Kiran; Jou, Jwo-Huei; **Vaidyanathan, Sivakumar***, “Unipolar 1-phenylimidazo[1,5-a]pyridine: A new class of ultra-bright sky blue emitters for solution-processed organic light emitting diodes” - *New J. Chem.*, **2022**, 46, 16717-16729, DOI:10.1039/D2NJ01938H, Advanced Article (**chosen for back cover**). (Citation – 6), Impact factor: 3.925 **Q2**
37. Jaipal Devesing Girase, Mangey Ram Nagar, Shah Nawaz, A. Choudhry, Jwo-Huei Jou and **Sivakumar Vaidyanathan***, “Highly Efficient Multifunctional luminogens for Near UV/Deep Blue (CIEy $\square$ 0.02) and Hybrid White OLEDs (CIE~0.33, 0.37) with Superior Color Stability” – *ACS Appl. Elec. Mater* **2022**, 4, 9, 4368–4382 DOI: [org/10.1021/acsaelm.2c00648](https://doi.org/10.1021/acsaelm.2c00648). (Citation – 5), Impact factor: 4.494 **Q1**
38. Jaipal Devesing Girase, Shah Nawaz, Jwo-Huei Jou, Sabita Patel and **Sivakumar Vaidyanathan***, “Solution-processed deep-blue fluorophores based on phenanthroimidazole integrated with benzimidazole with HLCT character for efficient deep-blue organic light emitting devices” - *Dyes and Pigments*, **2022**, 206, 110623 DOI: [org/10.1016/j.dyepig.2022.110623](https://doi.org/10.1016/j.dyepig.2022.110623). (Citation – 15), Impact factor: 5.122 **Q1**
39. Sibani Mund, K. Singh, M. Panda, B. Biswal, U. Subuddhi and **Sivakumar Vaidyanathan*** “Thiabendazole: a new class of antenna core structure for multifunctional trivalent organo-europium (EuIII) complexes ” - *J. Mater. Chem. C*, **2022**, 10, 10645 – 10659, Advance Article, DOI: [org/10.1039/D2TC01078J](https://doi.org/10.1039/D2TC01078J), (Citation – 7), Impact factor: 8.067 **Q1**
40. Md Intekhab Alam, Mangey Ram Nagar, Sandhya Rani Nayak, A. Choudhry, Jwo-Huei Jou and **Sivakumar Vaidyanathan***, "Acceptor Interlocked Molecular Design for Solution-Processed Stable Deep-Blue TADF and Hyper Fluorescence Organic

- LED Enabling High-efficiency" - *Adv. Opt. Mater.*, **2022**, 10 (18), 2200376. DOI: [org/10.1002/adom.202200376](https://doi.org/10.1002/adom.202200376). (Citation – 19), Impact factor: 10.05 **Q1**
41. Kasturi Singh, and **Sivakumar Vaidyanathan*** “Stable and efficient narrow band red emitters with high colour purity for white LEDs and plant growth applications” - *Dalton Transactions*, **2022**, DOI: 10.1039/D2DT01042A, **(appeared on back cover)** (Citation – 11), Impact factor: 4.569 **Q1**
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29. **V. Sivakumar** and U.V. Varadaraju (2005) Photoluminescent properties of NaY(WO₄)_{2-x}(MoO₄)_x:Eu³⁺ phosphor. *National Conference on Luminescence and its Applications*, Bangalore, India.
30. **V. Sivakumar** and U.V. Varadaraju, “Luminescence of Sm³⁺ in Eulytine, Ba₃La(PO₄)₃” in Chemist’s Meet – 2004, March 2004, Organized by Chemistry Department, IIT Madras, India.
31. **V. Sivakumar**, N. Lakshminarasimhan and U.V. Varadaraju, “Synthesis and Optical Properties of Calcium Substituted Monazite CePO₄” in Materials for Future (MF 2004), Feb 2004, Organized by Materials Research Centre, IISC Bangalore, India.

Annexure “C”

Ph.D Students completed/Ongoing:

S. No	Name of the student	Thesis title	Date of joining	Status	Current affiliation
1.	Boddula Rajamouli	Luminescent Eu ^{III} complexes based on phenanthro-imidazole ligands for white LEDs/OLEDs and temperature sensors: Combined experimental and theoretical investigations	Jan 2013	Graduated on Oct 2017	Associate Professor and Assistant Dean (Research) Department of Chemistry, School of Sciences and Humanities, SR University, Ananthasagar, Hasanparthy

					Warangal - 506371, Telangana, India.
2.	Kajjam Aravind Babu	Design, synthesis, photophysical and theoretical investigation of Iridium (III) complexes and organic luminophores for LEDs/OLEDs and chemosensors	Jan 2013	Graduated on April 2018	Post-doctoral researcher at Wayne State University, USA
3.	Mrs. S. Kasturi	Exploration of new lanthanide based inorganic phosphors intended for light emitting diodes	July 2013	Graduated on July 2019	WISE-PDF
4.	Tagare Jairam	Rational design, synthesis, photophysical and electroluminescence study of organic fluorophores for blue organic light emitting diodes	June 2014	Graduated on Jan 2020	Post-doctoral Researcher at Gachon University, South Korea
5.	Mrs. Rachna Devi	Exploration of New Eu(III) Complexes and their Applications in Light Emitting Diodes and Sensors: Combined Experimental and Theoretical Investigations	Jan 2016	Graduated on Aug 2021	Working in Honeywell New Delhi
6.	R. Marikumar	New Lanthanide based Inorganic Phosphors Intended for Solid State Lighting-Structure-Property-Correlations	Dec 2015	Graduated on Dec 2021	Assistant Professor, Department of Chemistry, UPL University of Sustainable Technology, Ankleshwar, Gujarat
7.	Mrs. Sibani Mund	Molecular Engineering of Efficient Light Harvesting Organic Antennas for Multitasking Organo EuIII Complexes: Design, Synthesis, and Photophysical Investigations	Nov 2022	March 2023	Lecturer at Municipal College Rourkela, Sundergarh Odisha
8.	Jaipal Devesing Girase	Molecular Engineering of Organic fluorophores for blue light emitting diodes: Synthesis, Photophysical and electroluminescence investigation	Sep 2022	April 2023	Postdoc - Gyeongsang National University, Jinju – South Korea
9.	Ms. Sandhya Rani Nayak	Molecular design, synthesis and photophysical investigations of blue emitting fluorophores for organic light emitting diodes and sensing.	Feb 2023	July 2023	NPfD at IISER Pune
10.	Md Intekhab Alam	Investigation of delayed emission from singlet and triplet states in pure organic molecules: design, synthesis, photophysical and electroluminescent applications	March 2023	June 2023	Post Doc at IISc Bangalore

11.	Priyadharsini Pradhan	Lanthanide based phosphors and their versatile applications	July 2019	Graduated in April 2025	
12.	Bhabana Priyadarshini Debata	Design and Synthesis of blue emitters for organic electronics	Jan 2021	Thesis submitted – Aug 2024	
13.	P. Sharma	Red-emitting phosphor and their applications	July 2018	Thesis submitted – Sep 2022	
14.	Swetha	Europium and Samarium complexes for smart LEDs and thermometers	Aug 2023	Ongoing	*presently at NIMS for the NIMS-IITH project
15.	Prathap R	TADF materials for OLEDs	Jan 2024	Ongoing	CSIR
16.	Biswaranjan Sahu	Fluorophores for Organic Electronics	Aug 2024	Ongoing	CSIR
17.	Pradyumna Sahoo	Fluorophores for Organic Electronics	Jan 2025	Ongoing	CSIR
18.	Sivakumar Velliyan	Red-NIR Phosphors	Jan 2025	Ongoing	CSIR
19.	Samatha Devineni	Lanthanide-based metal complexes for LEDs and Thermometers	Jan 2025	Ongoing	WISE-PhD
20.	Sajeetha Parveen Banu U	Red-NIR Phosphors	July 2025	Ongoing	MOE

M.Sc Students: Completed:

S. No	Name of the student	Thesis title	Date of completion	Current position
1.	S. Kasturi	Analysis of red muck from optimal by product gas distribution in the iron and steel making process	May 2013	Research Associate at NIT RKL
2.	Santanu Dalai	Synthesis and photoluminescence studies on Eu^{3+} - substituted double Perovskites	May 2013	Asst. Engineer, Tata Steel Pvt. Ltd.,
3.	Suraj Kumar Panigrahi	Synthesis and Photophysical studies of Bipolar Host materials for Phosphorescent OLEDs	May 2014	Lecturer
4.	Ajit Kumar Pillai	Design and synthesis of new multifunctional europium molecular complex for OLEDs	May 2014	Lecturer, Govt Polytechnique, Odisha

5.	Nikhil Kumar	Synthesis, characterization and spectral studies of novel Eu ³⁺ based phosphors and its solid solutions	May 2015	Assistant Professor, Aakash Educational Services Limited
6.	Onkar Kumar Das	Lanthanide based Phosphor materials for White LEDs	May 2015	Asst. Engineer, Jindal Steel Pvt. Ltd.,
7.	Rahul	Synthesis and Photoluminescence study of Eu ³⁺ doped in mixed metal oxides	May 2016	Senior Lecturer, Fiitjee Limited
8.	R.V. Varun	Synthesis, Photophysical, electrochemical and theoretical investigation of Acenaphtene based Bipolar host materials for OLED Application	Jan 2016	Graduate student at UC Riverside
9.	Shraddha M	Synthesis and Characterization of phosphate based Eu materials for LED applications	Jan 2016	-
10	Priti prama	Synthesis of Tungstate based phosphor materials for LEDs	May 2018	-
11	Nikhil Verma	Triphenylamine based bipolar host materials for OLEDs	May 2018	Instructor – Akash Institute – Dehradun
12	Raksha Pal	Europium based molecular complexes for LEDs	May 2019	PhD Students at South Korea
13	Satish K Samal	Perovskite quantum dots	May 2019	PhD student at INST Mohali
14	Savita Priya	Inorganic phosphors for solid state lighting	June 2020	PhD student at IIT Bombay
15	Sivaji Nayak	Red phosphor with layered structure for white LEDs	June 2020	
16	Siddhartha Sankar Barman (419CY2039)	Room temperature phosphorescence organic materials	Jun 2021	
17	Deepak Seth (419CY2057)	Blue emitting fluorophores for OLEDs	Jun 2021	
18	Harapriya Behera	Room temperature phosphorescence organic materials	Jun 2021	

19	Vimal Narendra Chauhan	Europium complexes for LEDs	Jun 2021	
20	Aayush Aryan	Phosphor materials for solid state lightings	June 2022	IITH
21	Sowbhick patra	Thermally activated delayed fluorescence organic materials	July 2022	
22	Satyajit biswas	Europium complexes for LEDs	July 2022	
23	Chawan Jyothi Singh	Novel red phosphors for SSL applications	July 2022	
24	Rushabh Nanji	Design and synthesis of TADF emitters for OLEDs	June 2024	PhD student at UK University
25	Sudipta Das	Phosphors for LEDs and thermometers	June 2024	
26	Sereya Roy	Design and synthesis of Europium complexes for white LEDs	June 2025	
27	Rohit	Phosphors for LEDs and thermometers	June 2025	

Ongoing:

S. No	Name of the student	Thesis title	Date of joining
1.	Apurv Uady R	Fluorophores for OLEDs	Ongoing
2.	Nishant M	Fluorophores for OLEDs	Ongoing

Intern Students: (completed)

S. No	Name of the student	Thesis title	Year
1.	Abhijit Mohanty	Eu ^{III} based complexes for OLEDs	Dec - 2015 & May – July 2016
2.	Sachidanada Nayak	Single molecular Eu complexes for white OLEDs	Dec - 2015 & May – July 2016
3.	Sonaleen Biswal	Triphenylamine based Donor- π -Acceptor organic phosphors for White LEDs	Dec - 2015 & May – July 2016

4.	Soumyasree Jena	Iminopyridine based Bipolar Host materials for OLED Applications	Dec - 2015 & May – July 2016
5.	Sathish Kumar Samal	Red emitting phosphor materials for LEDs	Dec - 2015 & May – July 2016
6.	Ipshita Padhy	Phosphor materials for LEDs	Dec - 2015 & May – July 2016
7.	Surabhi Sushree Nayak	Triphenylamine based Donor- π -Acceptor organic phosphors for White LEDs	Dec – 2015
8.	Ram Sankar Panigrahy	Bipolar host materials for OLEDs	Dec - 2015 & May – July 2016
9.	Anuj Kumar Ray	Star-shaped bipolar host materials for OLEDs	Dec - 2015

M.Sc and Integrated M.Sc Students (Completed):

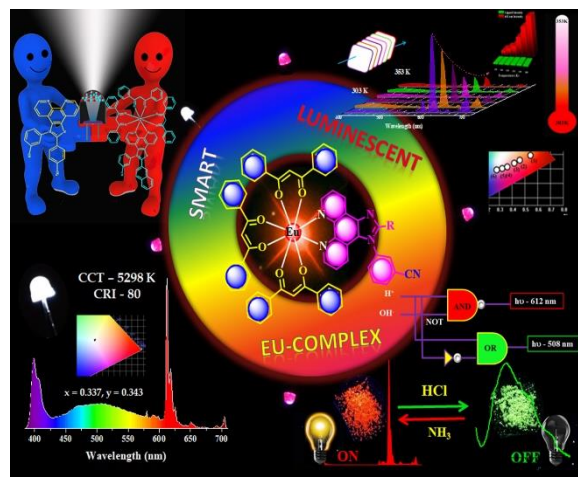
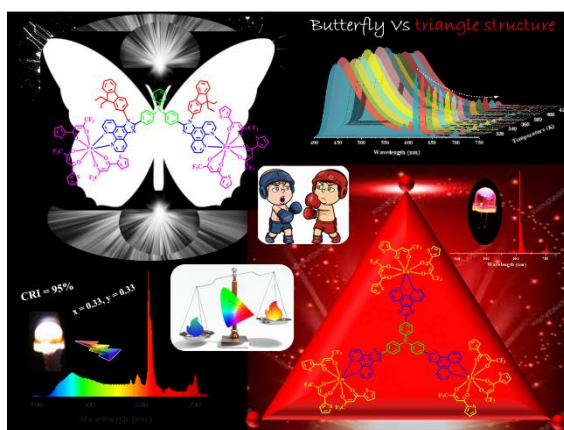
S. No	Name of the student	Thesis title	Year
1	Sadhana Kundu 1st year PG student Bit Mesra (Ranchi)	Synthesis and photophysical studies of organic fluorophores for chemo sensors	Dec 2017 – Jan 2018
2	Samikshya Sundarray II year MSc student Centurion university, Jatni, Bhubaneswar	Synthesis and photophysical studies of europium complexes for light emitting diodes	Feb – March 2018
3	Manini Kabi II year MSc student Centurion university, Jatni, Bhubaneswar	Synthesis and photophysical studies of organic blue fluorophores for OLEDs	Feb – March 2018
4	Spandan kumar mohanty, II year MSc student Centurion university, Jatni, Bhubaneswar	Synthesis and photophysical studies of organic fluorophores for chemo sensors	Feb – March 2018
5	Asima swain, I Year M.Sc Student, IGIT Sarang ,Dhenkanal	Synthesis and photophysical studies of blue organic fluorophores for LEDs	May 2018 – July 2018
6	Pooja Das, I Year M.Sc Student, IGIT Sarang ,Dhenkanal	Synthesis and photophysical studies of blue organic fluorophores for LEDs	May 2018 – July 2018

7.	Bijayalaxmi Routray II year MSc student Centurion university, Jatni, Bhubaneswar	Synthesis and photophysical studies of blue organic fluorophores for LEDs	Jan 2019 – Feb 2019
8.	Subhalaxmi Swain	Synthesis and photophysical studies of blue organic fluorophores for LEDs	Jan 2019 – Feb 2019
9.	Kiran Biswal	Synthesis and photophysical studies of Eu ^{III} complexes for LEDs	Jan 2019 – Feb 2019
10.	Priyanka Das	Synthesis and photophysical studies of Eu ^{III} complexes for LEDs	Jan 2019 – Feb 2019
11.	Subhasmita Nayak	Phosphor materials for solid-state lighting applications	Jan 2019 – Feb 2019
12.	Anwasha Pattanaik	Phosphor materials for solid state lighting applications	Jan 2019 – Feb 2019
13.	Aman Aryatosh IISER Kolkata	RTP fluorophores	May – July 2023
14.	Mithan Kumar G R IISER Tirupati	RTP fluorophores	May – July 2023
15.	Navya Vijayan NIT Calicut	Single component white Light emissive fluorophores	May – July 2023
16.	Ipsite Mishra Sambalpur University Odisha	Single-component white light- emissive fluorophores	May – July 2023
17.	Sathwika R NIT Warangal	Phosphor Materials for LEDs	May – July 2023
18.	Deekshit	Phosphor Materials for LEDs	May – July 2023
19.	NIT Warangal	Phosphor Materials for LEDs	May – July 2023

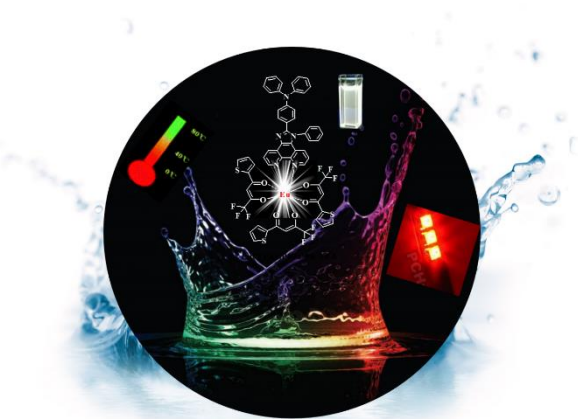
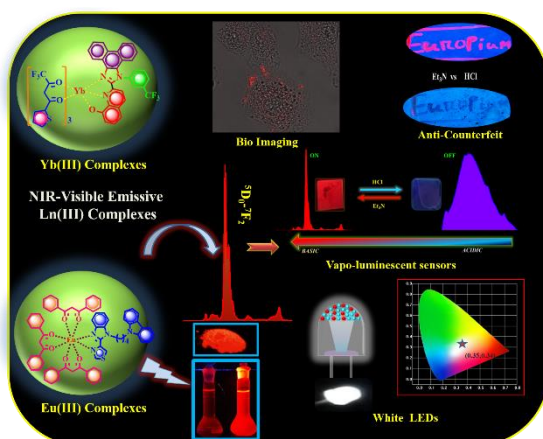
Annexure “D”

Current research activities at IITH:

1. Lanthanide-containing Molecular complexes for OLEDs, sensors, and White LEDs:
Molecular design and synthesis of Europium (Eu) and Dysprosium (Dy) containing molecular complexes and exploring the same use as light-emitting molecular systems for OLEDs/LEDs.



Very recently, we have synthesised Red-NIR emitting lanthanide complexes and study their applications in white LEDs, chemosensors/ratiometric temperature as well as bio-imaging.



2. New highly efficient Rare earth ion activated phosphor materials for solid-state lighting (SSL): To find alternative red phosphor for nitrides and sulfides, we explored numerous Eu^{3+} activated oxide phosphor materials with different crystal structures (including perovskite, scheelite, etc.). In addition, we are also working on Eu^{2+} and Ce^{3+} luminescence in robust host lattice for creating single host white light emission and color tuning emission.

Heavy metal ion-free persistent room temperature phosphorescence (RTP) of pure organic materials is drawing continued attention, due to their emerging applications in anticounterfeiting, bio-imaging, encryption, sensing, and optoelectronics. Compared to inorganic counterparts (organometallic complexes or lanthanide-doped or undoped persistent phosphors), organic RTP materials showed promising extended emission PL lifetime, significant Stokes' shift, rich excited state photophysical properties, and easy modification/fabrication. However, design and synthesis of pure organic room temperature phosphorescence (RTP) luminogens with simultaneous realization of ultra-long lifetime under ambient conditions and high efficiency remains a formidable challenge, owing to their conflicting requirements for the T_1 nature of (n,π^*) and (π,π^*) characteristics, respectively. We have recently synthesized a carbazole-based donor-acceptor system, which showed enhanced PLQY and lifetime.

